Worksheet 4.7

Electron configurations

Answers

Complete the following tables.

Name	Symbol	Atomic number	Electron configuration
Hydrogen	H	1	1
Helium	He	2	2
Lithium	Li	3	2, 1
Beryllium	Be	4	2, 2
Boron	B	5	2, 3
Carbon	C	6	2, 4
Nitrogen	N	7	2, 5
Oxygen	O	8	2, 6
Fluorine	F	9	2, 7
Neon	Ne	10	2, 8
Sodium	Na	11	2, 8, 1
Magnesium	Mg	12	2, 8, 2
Aluminium	Al	13	2, 8, 3
Silicon	Si	14	2, 8, 4
Phosphorus	P	15	2, 8, 5
Sulfur	S	16	2, 8, 6
Chlorine	Cl	17	2, 8, 7
Argon	Ar	18	2, 8, 8
Potassium	K	19	2, 8, 8, 1
Calcium	Ca	20	2, 8, 8, 2

Ion	Symbol	Atomic number	Number of electrons	Electron configuration	Matches configuration of
Cations					
Lithium	Li^+	3	2	2	Helium
Beryllium	Be^{2+}	4	4	2	Helium
Boron	B^{3+}	5	5	2	Helium
Sodium	Na^+	11	10	2, 8	Neon
Magnesium	Mg^{2+}	12	12	2, 8	Neon
Aluminium	Al^{3+}	13	10	2, 8	Neon
Potassium	K^+	19	18	2, 8, 8	Argon
Calcium	Ca^{2+}	20	18	2, 8, 8	Argon
Anions					
Nitride	N^{3-}	7	10	2, 8	Neon
Oxide	O^{2-}	8	10	2, 8	Neon
Fluoride	F^-	9	10	2, 8	Neon
Phosphide	P^{3-}	15	18	2, 8, 8	Argon
Sulfide	S^{2-}	16	18	2, 8, 8	Argon
Chloride	Cl^-	17	18	2, 8, 8	Argon

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